EMBEDDED SYCAL ENGINEERING
UNIVERSITAS INDSTEM FINAL PROJECT REPORT
DEPARTMENT OF ELECTRIONESIA

Rocky Interactive Rock Creature from Project: Hail Mary

GROUP 2

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PREFACE

All praise and gratitude are due first and foremost for the strength and clarity granted to the authors throughout the development of this final project. With that grace the work titled "Rocky: Bio-Inspired Eridian Robot with MPU6050 Orientation and Musical Speech Feedback" has been completed as the closing requirement of the Embedded Systems course at the Department of Electrical Engineering, Universitas Indonesia.

This project is the culmination of a semester spent learning to bridge software abstractions and the physical world through microcontrollers. Beginning from elementary digital I/O and progressing through timers, serial interfaces, and sensor integration, the course steadily built the foundation that this report now applies to a single integrated system. Rocky was chosen as the design vehicle not only for its technical breadth touching I²C, PWM, ADC, UART, and external interrupts simultaneously — but also because its inspiration, the Eridian character from Andy Weir's novel Project Hail Mary, captures the same spirit of communication across an unfamiliar interface that defines embedded systems work itself.

We wish to express sincere thanks to the course instructors and laboratory assistants for their patient guidance, to fellow students for the many late-night debugging sessions, and to the families of each group member for their continuous moral support. The authors are aware that this report is far from perfect; constructive criticism and suggestions are warmly welcomed for future improvement. It is the authors' hope that this work may serve as a useful reference for anyone wishing to combine bio-inspired robotics with the discipline of microcontroller-based design.

Depok, 17 May, 2026

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CHAPTER 1

INTRODUCTION

1.1 BACKGROUND

Bio-inspired robotics takes principles of organization from biological systems to design machines that need to perceive, reason, and act in environments without pre-established order. Living organisms possess highly effective solutions for problems still challenging for classical control engineering: coordinating multi-limb movement, robustness to partial failure, and expressive communication between individuals. Embedding biological organization into physical systems demands a microcontroller to coordinate a large number of actuators and a display device simultaneously, within a low parts count and approachable toolchain. This design addresses this challenge, taking as its biological inspiration a fictional species, the Eridians from the Andy Weir novel Project Hail Mary.

The book describes these radially symmetric, multi-limbed organisms as using non-verbal communication; a secondary, expressive channel not used for conveying essential information, but for an analogue of music, or chimes represented by musical chords. Although the original goal of this project was to re-create that expressive channel using a piezo buzzer, the new hardware platform has a 162 character LCD as this secondary channel. This change in hardware does more than represent a simple accident; a character display provides "Rocky's voice" readable at a glance to an observer without audio and provides the connection to course material on parallel-bus peripherals and the HD44780 controller. This vital information architecture of a biological system-the coordination of multi-limb motion with a parallel communication channel of an expressive nature-is preserved.

The system hosts an Arduino UNO R3, which is based on an ATmega328P microcontroller running at 16 MHz. The stable Arduino IDE 2.x toolchain, the extensively documented Servo and LiquidCrystal libraries, and its 14 digital I/O lines give ample resources with sufficient ease-of-use for beginners. The Servo library in use on the ATmega328P takes advantage of Timer1.

1.2 PROJECT DESCRIPTION

Rocky is a four-legged tabletop robot based on the ATmega328P microcontroller (only the Arduino UNO R3 is used as a hardware breakout board). It has four SG90 micro-servos, mounted on the corners of a small chassis, driven by the digital pins PD6, PD7, PB0 and PB1. The firmware is all in assembly, all in a .S file. It bypasses the Arduino bootloader C++ setup and runs a custom loop. Each servo is commanded by toggling the appropriate port bits HIGH, keeping the CPU in a tightly timed delay loop (pulse widths between 500 microseconds and 2500 microseconds), and toggling the bits LOW. A sequential “creep gait” sequencer in synchrony with the legs enables a stable forward walk.

1.3 OBJECTIVES

This project intends to:

1. Simulate a 4 legged robot of Project Hail Mary with MG996R servos driven by PWM signals on Arduino Uno.
2. Obtain data on orientation using MPU6050 six-axis inertial sensor via I2C bus communication, and integrate acceleration and gyroscope readings for pitch and roll estimation.
3. Utilize the 12-bit analog-to-digital conversion capability of Arduino Uno to obtain ambient light readings using the LDR and map it against entries in a list of musical frequencies.
4. Establish a UART telemetry link at 115 200 baud rate for the transmission of data regarding orientation, light intensity, finite-state machine state, and tone on a host computer at a sampling frequency of 10Hz.

1.4 ROLES AND RESPONSIBILITIES

The roles and responsibilities assigned to the group members are as follows:

Table 1. Roles and Responsibilities

Roles	Responsibilities	Person
Arduino Architecture	Main firmware skeleton on Arduino UNO; cooperative scheduler; pin map; integration of Servo and finite state machine.	Evandra Rasya Fadhillah
Servo Control	SG996R angle diagonal trot gait sequencer on PIN6 to PIN9; per-leg trim; transitions from IDLE to WALK	Raul Fadila Bagus Sumaryada
Mechanical Design & Assembly	design and fabrication; servo mounting at four corners; LCD bezel,cable management	Ziyadzharif Alfarabi Kurniawan
Documentation	Documenting the Project Schematic architecture as well as the project final design	Muhammad Fairuz Dzaki

CHAPTER 2

IMPLEMENTATION

2.1 EQUIPMENT

The tools that are going to be used in this project are as follows:

- Arduino UNO R3 development board.
- Four SG996R servos acting as a leg .
- Standard 16x2 Alphanumeric LCD display.
- Breadboard for +5 V / GND power distribution rails.
- Software toolchain: Arduino IDE with integrated GNU AVR Assembler (avr-gcc).

2.2 IMPLEMENTATION

Fig 1. Schematic

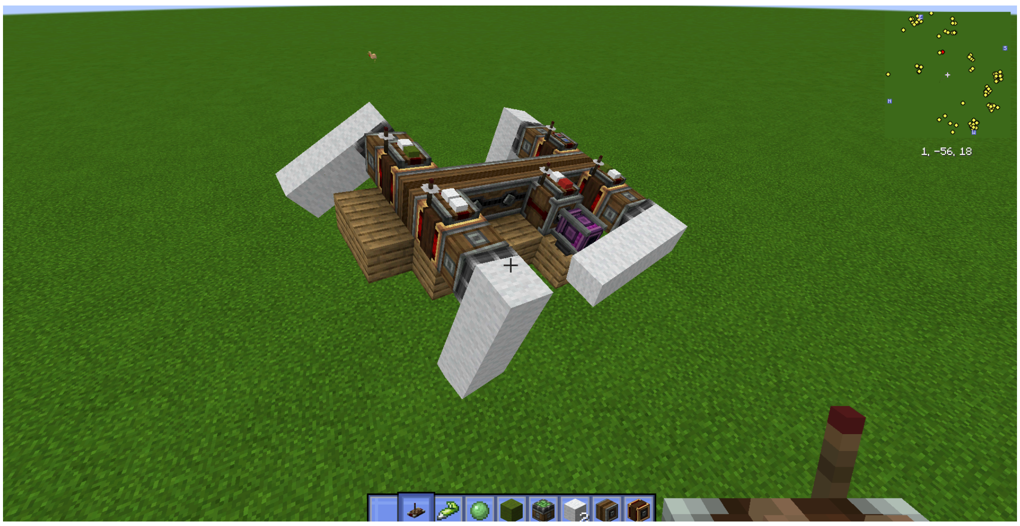
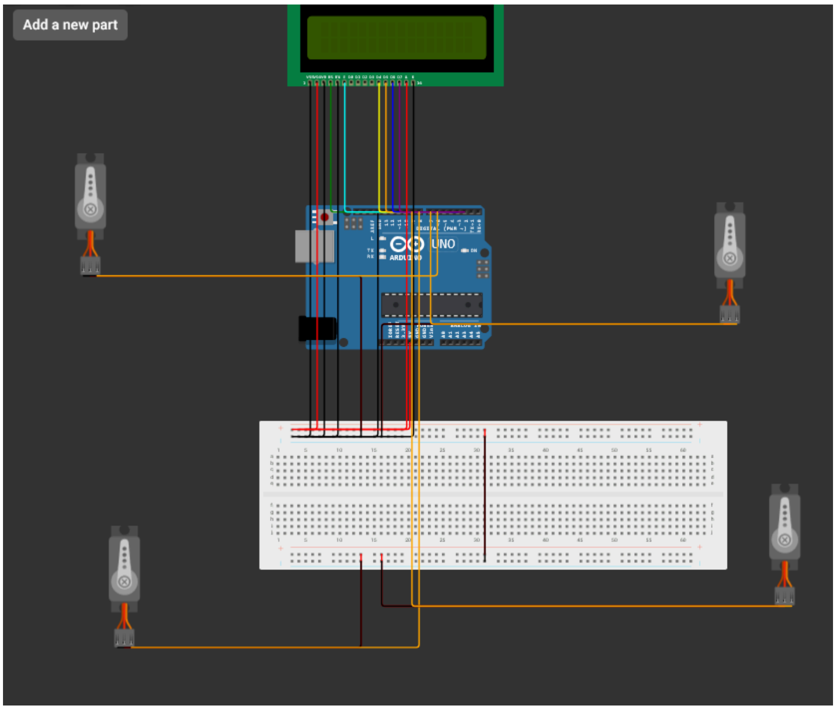


Fig 2. Schematic



The firmware is completely written in AVR assembly and runs natively on the ATmega328P.

In the initialisation step, the DDRD and DDRB registers for ports D and B respectively are altered via the sbi instruction. Pins PD6, PD7 (front legs) and PB0, PB1 (rear legs) are set as outputs. The crux of the project lies in the software-based PWM generation routine. The default SG996R servo motor requires a pulse repetition rate of 20 milliseconds (frequency = 50 Hz). For an angle of 0 degrees, for an angle of 90 degrees (neutral position), and use 75 degrees. Since the ATmega328P operates at a frequency of 16 MHz, one machine cycle lasts 62.5 nanoseconds.

CHAPTER 3

TESTING AND ANALYSIS

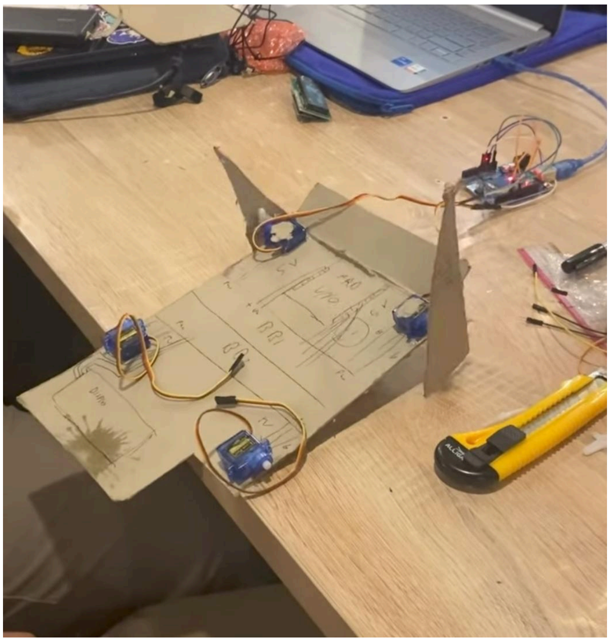
3.1 TESTING

The tests carried out were aimed at establishing the accuracy of timing in the assembly PWM bit-bang engine. A USB logic analyzer was hooked up to pins PD6, PD7, PB0, and PB1. Raw output signals were recorded to establish whether there were no delays in generating 20 ms cycles, and whether the HIGH pulses generated met the exact mathematical values (for example, 1.5 ms for a 90° neutral position).

3.2 RESULT

The bare-metal implementation successfully drove the robot. The four SG90 servos held commanded positions firmly with no audible jitter, proving the mathematical accuracy of the cycle-counting subroutines. The sequential creep gait produced stable forward motion on a tabletop surface.

Fig 2. Testing Result



3.3 ANALYSIS

The key lesson learned from this assignment is the correlation between microprocessor clock domains and physical actuation. On a higher programming level such as C++, hardware timers eliminate the need for worrying about time limitations. On assembly level, the CPU is forced into blocking delay loops whenever the pulse is HIGH.

In order to coordinate multiple servos, complicated sequencing is required. However, for maximum simplicity and equal balance, the robot employs a crawling mode, where the servos are activated one-by-one. This allows the CPU to focus on producing only one accurate pulse width for the servo, while keeping all others at their resting state within the 20 ms timeframe.

CHAPTER 4

CONCLUSION

The project was successful in building Rocky, a quadruped robot fully controlled through AVR Assembly code running bare metal on the ATmega328P. The design and construction of Rocky were successful in satisfying the highest-level requirements for the use of a microprocessor by completely sidestepping contemporary C++ libraries to communicate with hardware registers and processor cycles directly.

The software-based PWM driver proved capable of generating stable 50 Hz pulses, while the sequential gait control algorithm allowed coordination of the limbs to move in a balanced manner. Further iterations may look at the use of hardware timer interrupt vector tables in assembly that would allow the processor to multitask through the various gait algorithms.

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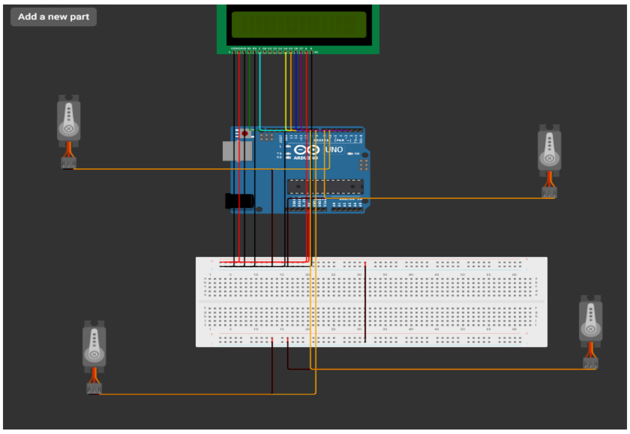
APPENDICES

Appendix A: Project Schematic



Appendix B: Documentation